

Reference Core: gem5 MinorCPU pipeline

Memory hierarchy

L1 instruction cache

Main memory (DRAM)

L1 data cache

Frontend

speculative regime

Fetch1

I-cache request

PC select / next-PC mux

Fetch2

32-bit cache line
→ instruction(s)

branch pre-decode

Branch predictor

BHT · BTB · RAS

direction / target / returns
consulted on each control inst

fetch request
/ cache line

prediction redirect (Fetch2 → PC select)

Decode

in-order

Decode

macro-op split

Execute

in-order issue, out-of-order completion

Issue (scoreboard)

single-issue, in-order

Functional units

Integer

IntALU

IntMul

IntDiv

Floating point

FPAdd/Cvt

FPMul

FPDiv

FPSqrt

memory ops via the LSQ

Branch resolution

Load / Store Queue

loads issued early

completed

Commit

non-speculative

Commit

in-order retire

load · store / data

Front-end

Back-end

resolution redirect (Execute → PC select)



decoupling FIFO: each stage's input buffer